

# Deep Learning for Solving Dynamic Models

Session 7 (exercise) — Surrogate-based structural estimation via SMM

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# The Model: Brock–Mirman with Stochastic TFP

Stochastic growth model:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \quad K_{t+1} + C_t = Y_t + (1 - \delta)K_t$$
$$Y_t = z_t K_t^\alpha, \quad \log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

**Calibrated parameters**

| $\alpha$ | $\delta$ | $\sigma$ | $\beta$ (single- $\varrho$ exercise) |
|----------|----------|----------|--------------------------------------|
| 0.36     | 0.10     | 0.04     | 0.96                                 |

Note: the textbook closed-form policy  $s = \alpha\beta$  requires  $\delta = 1$ . We use  $\delta = 0.10$ , so no analytic policy exists and the Euler equation must be solved numerically – the surrogate replaces this expensive inner solve.

**Pedagogical sequence:**

- ▶ **Exercise 1:** estimate  $\varrho \in [0.50, 0.99]$  from autocorrelation moments. Sharp 1D identification.
- ▶ **Exercise 2:** estimate  $(\beta, \varrho)$  *jointly*. Visualize identification ridge; collapse it with extra moments.

# Why $\varrho$ , not $\beta$ ?

- ▶  $\beta$  is famously **poorly identified** from macro moments alone – it shifts the level of the savings rate but every level moment depends on the same scalar.
- ▶ In practice,  $\beta$  is calibrated to a target steady-state interest rate (under log utility and full depreciation,  $1/\beta = 1 + r$  at SS; with general  $\delta$ , the SS Euler relation is  $1/\beta = 1 + \alpha(k^*)^{\alpha-1} - \delta$ ).
- ▶  $\varrho$  has a **clean direct mapping** to autocorrelation moments:  $\text{Corr}(\log Y_t, \log Y_{t-1})$ ,  $\text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$ , and the simulated volatility moment move with persistence.
- ▶ This is the textbook example of a **sharply identified** scalar parameter – exactly what we want for teaching the SMM workflow. (With three moments and one parameter the system is technically *over-identified*; the extra moments make the criterion sharper.)

The single-parameter exercise (notebook 03) recovers  $\hat{\varrho}$  accurately under CRN. The joint exercise (notebook 03b) shows what happens when an additional, partially-identified parameter is freed.

# Task 1: Train Surrogate with $\varrho$ as Pseudo-State

## Goal

Train  $\mathcal{N}(z_t, K_t, \varrho) \rightarrow s_t \in (0, 1)$ , with  $K_{t+1} = (1 - \delta)K_t + Y_t \cdot s_t$ .

1. 3-input MLP  $(z, K, \varrho) \rightarrow s$  with sigmoid output.
2. Euler residual loss with Gauss–Hermite quadrature;  $\varrho$  enters the *transition* as well as the policy. Under log utility,  $u'(C) = 1/C$ , so the Euler ratio simplifies to  $u'(C_t)/u'(C_{t+1}) = C_{t+1}/C_t$ , and dividing by  $C_t$  gives:

$$\mathcal{L} = \frac{1}{N} \sum_i \left( 1 - \beta C_t^{(i)} \sum_j w_j \frac{1 - \delta + \alpha z_{t+1,j}^{(i)} K_{t+1}^{(i)\alpha-1}}{C_{t+1,j}^{(i)}} \right)^2$$

3. Two-phase training: (i) uniform sampling on  $(z, K, \varrho)$  box, (ii) simulation-based refinement.
4. Validate by plotting  $s(z = 1, K, \varrho)$  for several  $\varrho$ .

# Task 2: Simulate and Compute Identifying Moments

## Goal

Generate synthetic “ex-post” data at  $\varrho_{\text{true}} = 0.90$  and compute three identifying moments (plus one diagnostic, see below).

| Moment                                                    | What it captures                                                      |
|-----------------------------------------------------------|-----------------------------------------------------------------------|
| $m_1 = \text{Std}(\Delta \log C_t)$                       | Growth-rate volatility; check numerically                             |
| $m_2 = \text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$ | Persistence in consumption growth                                     |
| $m_3 = \text{Corr}(\log Y_t, \log Y_{t-1})$               | Persistence in output (very direct)                                   |
| $m_0 = \text{mean}(s_t)$ ( <i>diagnostic only</i> )       | Should be flat in $\varrho$ ; included to show what <i>not</i> to use |

**Note:**  $\text{Var}(\log z) = \sigma^2/(1 - \varrho^2)$ , but  $\text{Var}(\Delta \log z) = 2\sigma^2/(1 + \varrho)$ .

**CRN:** same burn-in, horizon, initial state, seed for every candidate  $\varrho$ .

# Task 3: Estimate $\varrho$ via SMM

SMM objective (single-parameter case,  $\theta = \varrho$ )

$$\hat{\varrho} = \arg \min_{\varrho} [m(\varrho) - \hat{m}^{\text{data}}]^\top W [m(\varrho) - \hat{m}^{\text{data}}], \quad W = I.$$

$W = I$  is the unweighted criterion – the cleanest pedagogical default. The efficient choice is  $W = S^{-1}$  where  $S = \text{Var}(\sqrt{T} \hat{m}^{\text{data}})$ ; with  $W = I$  the asymptotic variance of  $\hat{\varrho}$  is suboptimal but consistency and the interior-solution argument are unchanged.

1. Define one routine  $\varrho \mapsto (C_t, I_t, Y_t) \mapsto m(\varrho)$  that reuses the trained surrogate.
2. Coarse grid pass on  $\varrho \in [0.50, 0.99]$  to localize the minimum.
3. Refine with `scipy.optimize.minimize_scalar` (bounded Brent's method).
4. **Verify the solution is interior** ( $\hat{\varrho}$  strictly inside the bounds), required for asymptotic theory.

## Hint

Re-seed the simulator every evaluation. Without CRN the criterion is dominated by simulation noise.

# Stacking a GP over the Moment Map (used in Part B)

**Forward reference.** Task 4 only needs the single- $\varrho$  surrogate; the construction below is what Notebook 03b adds for the *joint*  $(\beta, \varrho)$  exercise.

**Two-layer surrogate** for high-throughput SMM:

- ▶ **Layer 1.** Pseudo-state DEQN  $\mathcal{N}_\rho(z, K, \theta) \mapsto$  savings rate
- ▶ **Layer 2.** One Gaussian process per moment,  $\hat{m}_j(\theta) \sim \text{GP}(0, k_j)$ , fit on a small design  $\{(\theta^{(i)}, m(\theta^{(i)}))\}_{i=1}^n$

**Cost of one design point.** One  $T$ -step forward simulation under  $\mathcal{N}_\rho$  + a moment computation  $\Rightarrow$  expensive but parallel.

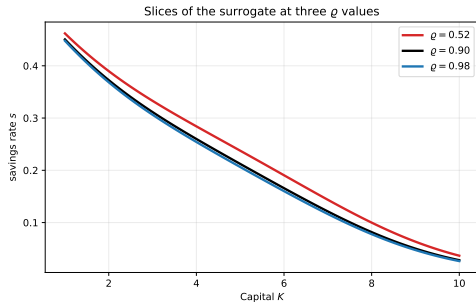
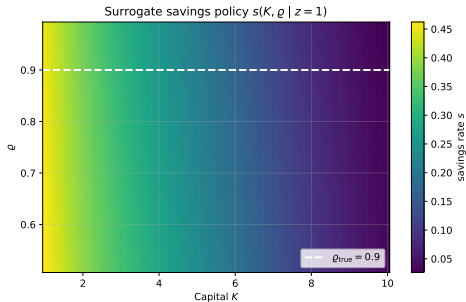
**Cost of one objective evaluation after training.** One GP posterior call per moment, no simulation. Bootstrap/SBI workflows then run on the GP.

**Diagnostic.** Cholesky-trick LOO on each moment GP (free, see deck 07). Notebook 03: 8 grid + 6 BAL points, fresh holdout for sanity.

# Task 4: Diagnose the Fit (Single-Parameter)

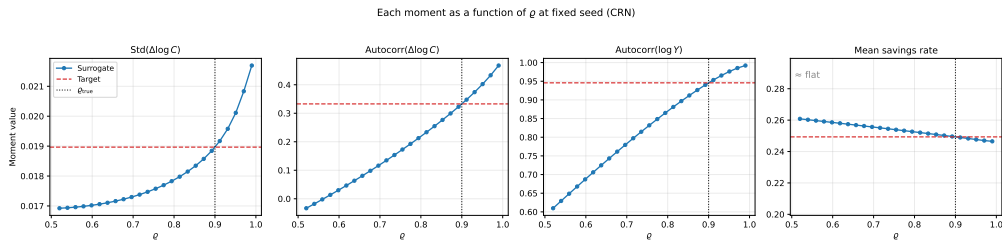
- ▶ How close is  $\hat{\varrho}$  to  $\varrho_{\text{true}} = 0.90$ ?
- ▶ Plot the SMM objective profile and mark both  $\varrho_{\text{true}}$  and  $\hat{\varrho}$ .
- ▶ Compare matched moments  $m(\hat{\varrho})$  versus targets  $\hat{m}^{\text{data}}$ .
- ▶ Plot  $s(K, \varrho_{\text{true}})$  vs.  $s(K, \hat{\varrho})$  – the curves should overlay almost exactly.
- ▶ Trace each candidate moment as a function of  $\varrho$  on the trained surrogate – this is the identification picture.

# Expected Result: Surrogate Slice $s(K, \varrho \mid z = 1)$



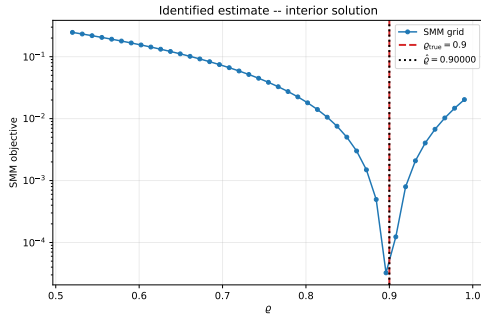
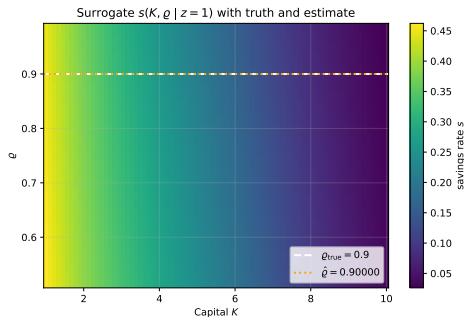
The 3-input network is a smooth function over the entire  $(K, \varrho)$  rectangle. The right panel shows three slices; the left panel shows the full surrogate compressed into one heatmap.

# Expected Result: Identifying Variation



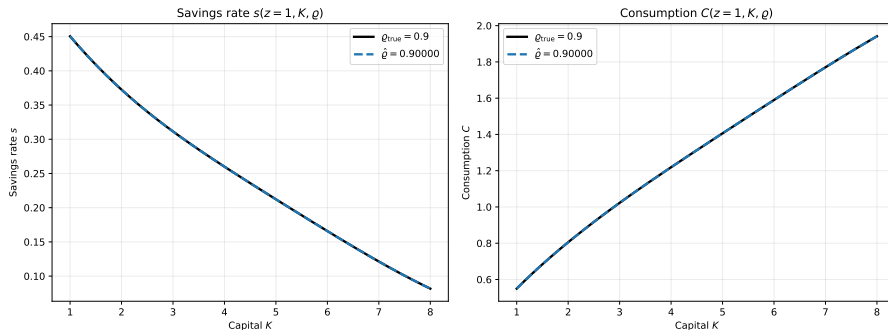
All four candidate moments traced as a function of  $\varrho$ : the three identifying moments  $m_1, m_2, m_3$  (Task 2 table, leftmost three panels) are monotone or near-monotone in  $\varrho$ . The diagnostic moment  $m_0 = \text{mean}(s_t)$  (rightmost panel) is essentially flat – correctly diagnosed as uninformative.

# Expected Result: SMM Objective + Estimate



Left: surrogate slice with truth (white dashed) and  $\hat{\varrho}$  (orange dotted). Right: log-SMM-objective profile with both. The estimate sits strictly inside the parameter rectangle (**interior** solution).

# Expected Result: Policy Match



The estimated policy at  $\hat{\varrho}$  overlaps the true policy at  $\varrho_{\text{true}} = 0.90$  to plotting accuracy. Single-parameter SMM with sharp identification recovers the truth accurately under CRN.

# Bridge: From Just-Identified to Joint

The single-parameter exercise is the cleanest version of the SMM workflow. But two real lessons remain:

- ▶ How does the surrogate scale to *multiple* parameters? Same idea: add them as pseudo-states.
- ▶ What does the SMM criterion look like when one parameter is **partially identified**?

Notebook 03b re-trains the surrogate on the rectangle  $\beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99]$  and estimates both jointly.

Two moment subsets:

- ▶ **Just-identified** (2 moments):  $\{ \text{mean savings}, \text{Corr}(\Delta \log C) \}$ .
- ▶ **Over-identified** (4 moments): adds  $\text{Std}(\Delta \log C)$  and  $\text{Corr}(\log Y)$ .

# Joint Task: Estimate $(\beta, \varrho)$

## Goal

$$\hat{\theta} = \arg \min_{\theta=(\beta, \varrho)} [m(\theta) - \hat{m}^{\text{data}}]^{\top} W [m(\theta) - \hat{m}^{\text{data}}] \quad \text{on } \beta \in [0.92, 0.99] \times \varrho \in [0.50, 0.99].$$

1. 4-input MLP  $(z, K, \beta, \varrho) \rightarrow s$ . Train once on the whole rectangle.
2. Forge synthetic data at  $(\beta_{\text{true}}, \varrho_{\text{true}}) = (0.96, 0.90)$ .
3. Compute the SMM criterion on a  $25 \times 25$  grid in  $(\beta, \varrho)$  – visualize both moment subsets.
4. Refine with `scipy.optimize.minimize` (L-BFGS-B, which honours the box constraints natively, started from the grid argmin).
5. Verify the solution is interior; report distance to truth on each axis.

# BoTorch UCB Acquisition for Joint Estimation

**Why active learning here.** On the 2D rectangle, most of the area sits far from the SMM minimiser – pure exploration is wasteful. We want a design that hits both the basin of attraction **and** regions where the moment GP is uncertain.

**Acquisition function** (notebook 03b):

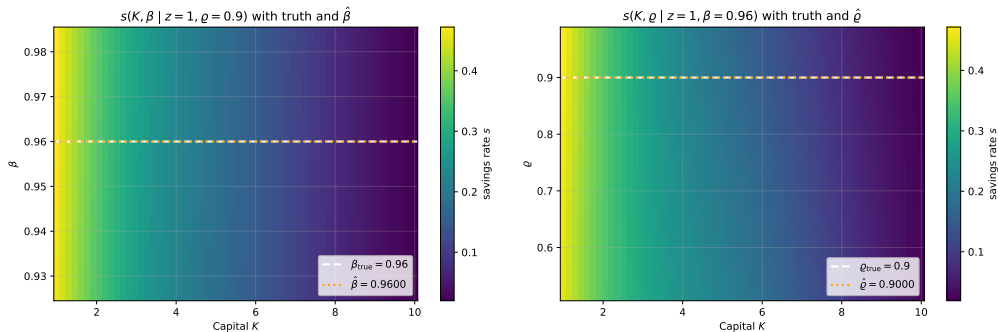
$$a(\theta) = [0.25 + \widetilde{\text{UCB}}_g(\theta)] \cdot \left\| \sigma_m(\theta) / \widehat{\text{sd}}(m) \right\|_2.$$

- ▶  $g(\theta) = -\log_{10}(Q(\theta) + \epsilon)$  is the negative log-criterion (a GP is fit to  $g$ );  $\widetilde{\text{UCB}}_g$  is BoTorch's standard  $\mu_g + \kappa\sigma_g$  rescaled to  $[0, 1]$  over the grid.
- ▶  $\sigma_m(\theta) \in \mathbb{R}^{n_m}$  stacks the moment-GP posterior std-deviations;  $\widehat{\text{sd}}(m)$  is the empirical std-deviation of each moment in the design (a unit-normaliser).
- ▶ 0.25: a small floor on the exploit factor so candidates with low UCB on  $g$  but high moment-GP uncertainty still enter the design.

**Roles.** First factor = *exploit* (basin of small  $Q$ ); second factor = *explore* (where moment GPs are uncertain). Pure UCB on  $g$  alone would over-collapse to the basin; pure moment-uncertainty would ignore the criterion shape.

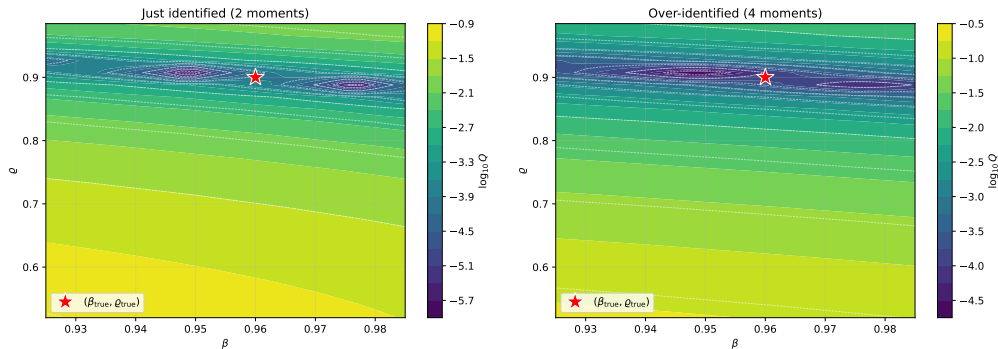
Built on BoTorch SingleTaskGP + UpperConfidenceBound; the multiplicative form is custom (not a stock BoTorch acquisition).

# Expected Result: Surrogate Slices on $(\beta, \varrho)$



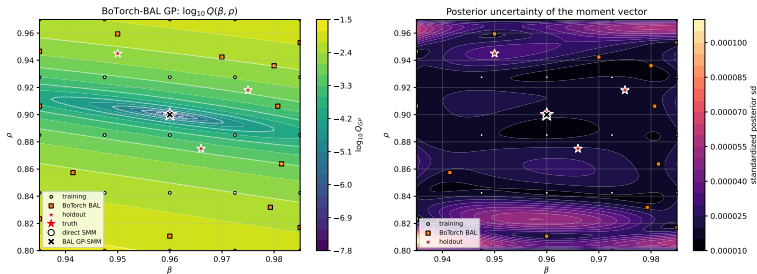
Two slices of the same 4-input network. White dashed = truth, orange dotted = over-identified estimate. The estimate sits strictly **inside** the rectangle on both axes simultaneously.

# Expected Result: 2D SMM Criterion



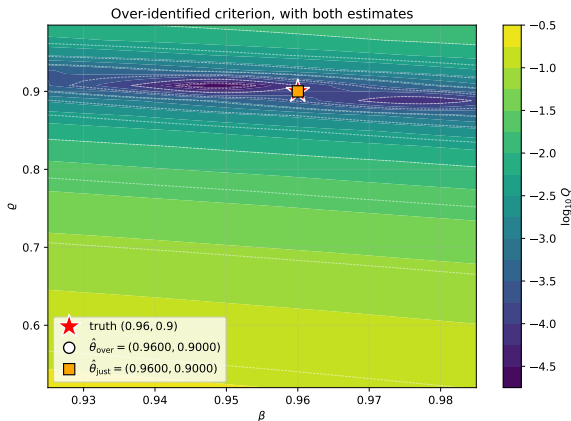
Left: the just-identified criterion shows a **ridge** along  $\beta$  – exactly the partial-identification problem that motivates calibrating  $\beta$  in practice. Right: adding two persistence moments collapses the ridge to a localized minimum.

# GP Surrogate of the SMM Criterion



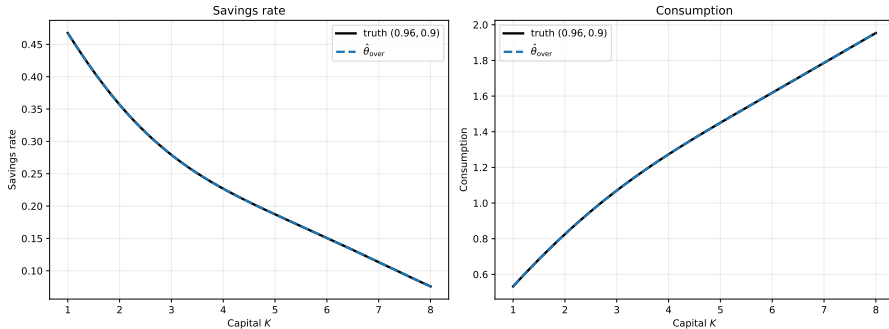
- ▶ Left: GP posterior mean of  $\log_{10} Q(\beta, \rho)$ ; right: standardized moment uncertainty.
- ▶ White dots = training design; orange squares = active additions; red stars = holdouts/truth.
- ▶ White circle = direct SMM; black cross = GP-SMM. Later objective calls need only one GP posterior evaluation.

# Expected Result: Estimates on Top of Criterion



In this synthetic CRN run, both estimators recover  $(\beta, \varrho)$  close to the truth. The takeaway is the criterion's *shape*: the just-identified ridge would translate into wide confidence bands on  $\beta$  in real-data applications, while the over-identified criterion has localized curvature on both axes.

# Joint Policy Match



Savings rate and consumption at the over-identified estimate overlay the truth almost exactly. The 4-input surrogate, trained *once*, supports millions of cheap moment evaluations during the optimizer's line-search.

# Wrap-Up

What the two notebooks teach together:

- ▶ A neural surrogate with parameters as pseudo-states **decouples** the cost of solving the structural model from the cost of moment-matching. Train once, use it inside any optimizer.
- ▶ For a well-identified parameter ( $\varrho$ ), SMM under CRN gives a sharp 1D identification picture.
- ▶ For **joint** estimation, the contour plot makes the partial-identification ridge visible. Adding moments collapses it.
- ▶ The **interior-solution** check is part of the standard pipeline, needed for the asymptotic distribution theory of the SMM estimator.

**Formal identification machinery** (moment Jacobian  $M = \partial m / \partial \theta$ , rank condition,  $J$ -statistic, weak-identification diagnostics): see lecture script Sec. 10.4 (Practical Considerations).

**Production pipeline:** Chen, Didisheim & Scheidegger (2026, *J. Financial Economics*), exactly this surrogate-and-SMM logic at scale, applied to finance.